

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain the different types of connectors used in communication and power transmission systems, along with their applications.

Q.20 Describe the different types of safety measures taken while handling cables and connectors, including the importance of proper clothing and protective gear.

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Roll No.

Level 5, 1st Sem / DVOC (Medical Imaging Tech.)

Subject : Tools, Equipment & Safety Measures - I

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

Q.1 What is the main purpose of an optical fiber cable?

- a) High-speed data transmission
- b) low-cost power distribution
- c) audio signal transmission
- d) grounding electrical systems

Q.2 Which of the following is a characteristic of a twin axial cable?

- a) it is used for ethernet networking
- b) it contains two conductors with a common axis
- c) it is used for heavy power distribution
- d) it has a single solid core

Q.3 Which of the following cables is commonly used for television signal transmission?

- a) twisted pair cable b) coaxial cable
- c) ribbon cable d) Optical Fiber Cable

Q.4 Which connector is commonly used for video output on computers?

- a) vga b) rj11
- c) usb d) bnc

Q.5 What is the function of a metallic sheathed cable?

- a) To protect the inner conductors from mechanical damage
- b) to improve signal transmission quality
- c) to reduce power consumption
- d) to make the cable lighter.

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 What are the advantages of using an armoured cable?

Q.7 Explain the use of USB connectors in electronic devices.

Q.8 What is the main advantage of twisted pair cables?

Q.9 Why is it important to wear proper ESD clothing while working with sensitive electronic components?

Q.10 What is the main difference between a multi-conductor cable and a twin-axial cable?

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Explain the structure and function of a ribbon cable.

Q.12 What are the safety precautions to be taken while handling electrical connectors?

Q.13 Describe the function of twin-lead cables and where they are used.

Q.14 What are the major advantages of shielded cables over unshielded cables?

Q.15 How does a coaxial cable differ from an optical fiber cable?

Q.16 Describe the working principle of an undergrounded power supply cable.

Q.17 How should ESD clothing be maintained for long-term use?

Q.18 What are the characteristics of a non-metallic sheathed cable?